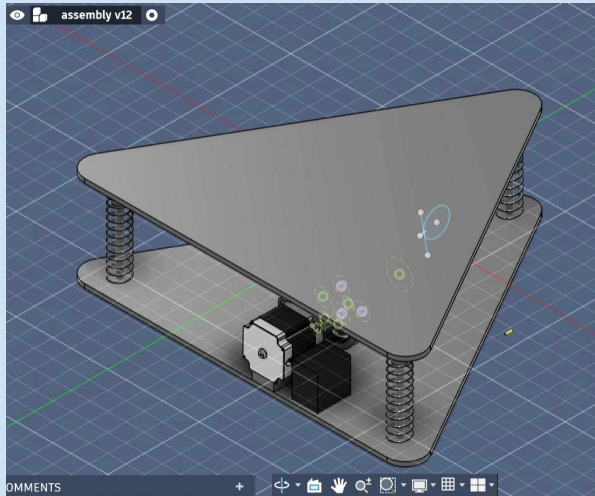


KineticStep

Turning Everyday Footsteps into Usable Energy



Every day, millions of people walk through metro stations, airports, railway stations, malls, universities, and other public spaces. These high-footfall areas have something in common: a constant flow of human movement. Every step taken on these surfaces produces mechanical energy, but this energy is currently lost.

KineticStep is designed to capture a part of this wasted energy and convert it into usable electricity.

Our solution is a modular kinetic floor tile that can be installed in areas with high and predictable pedestrian traffic. The electricity generated can be used locally to power low-energy applications such as LED lighting,

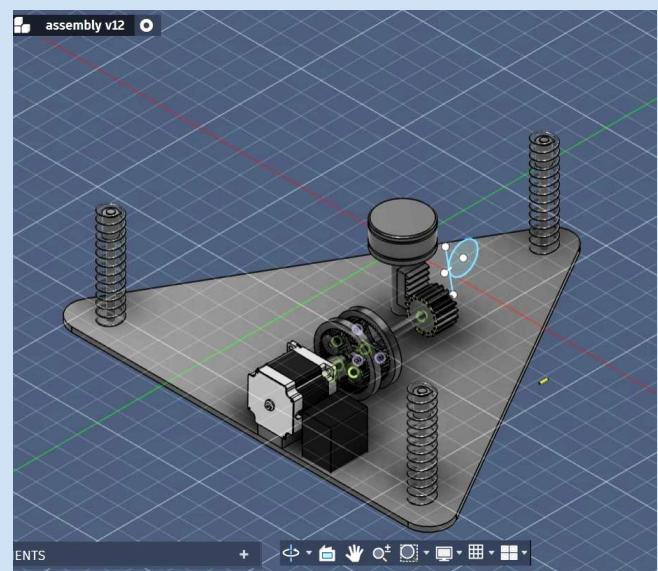
smart sensors, information displays, IoT devices, emergency lighting, and other connected infrastructure.

Beyond energy generation, KineticStep can also collect anonymized pedestrian movement data. This allows the same infrastructure to provide useful insights into footfall and crowd movement, creating a second potential revenue stream.

How KineticStep Works

When a person steps on the tile, the top surface moves vertically by approximately 15–20 mm. The movement is controlled using a guided compression mechanism with springs, bearings, and hardened steel guide rods, ensuring that the tile remains stable and comfortable to walk on.

This downward movement drives a rack-and-pinion mechanism, which converts the linear movement of the tile into rotational motion. The rotational motion then passes through a two-stage split-shaft planetary gearbox. Footsteps produce relatively slow rotational movement, so the gearbox increases the rotational speed by



approximately 25 times, allowing the generator to operate more efficiently.

The high-speed output drives a stepper motor functioning as a generator. The electricity produced is then converted into usable DC power through a rectification and charging circuit and stored in batteries. The stored energy can then be used immediately or supplied to nearby low-power systems.

Why KineticStep?

KineticStep is designed specifically for environments where people are already walking in large numbers. Instead of requiring additional land or changing people's behaviour, the system works by integrating energy harvesting directly into existing pedestrian infrastructure.

1. Works Where Solar and Wind Cannot

KineticStep does not depend on sunlight, wind, or weather conditions. It can operate:

- Indoors
- Underground
- At night
- In enclosed transport hubs
- In areas where solar installation is difficult

This makes it particularly relevant for metro stations, airports, railway terminals, and other dense urban environments.

2. Makes Use of Existing Infrastructure

The system is designed as a modular floor tile, allowing installations to be scaled according to the requirements of a location. A small installation can be used for a pilot project or a specific application, while multiple tiles can be connected across larger high-footfall areas.

3. Local Energy Generation

The electricity generated can be used close to where it is produced. This makes it suitable for powering applications such as:

- LED and pathway lighting
- Emergency lighting
- Smart sensors
- IoT devices
- Information displays
- Environmental monitoring systems
- Other low-power infrastructure

KineticStep is not intended to replace the electrical grid. Instead, it provides a way to recover energy that would otherwise be wasted and use it for nearby applications.

4. A More Durable Mechanical Approach

Many existing kinetic flooring systems rely on piezoelectric technology, which can be expensive and difficult to scale. KineticStep takes a different approach by using a mechanical transmission system built around a rack-and-pinion mechanism and a custom planetary gearbox.

Our goal is to develop a system that is simpler to manufacture, easier to maintain, and more cost-effective to deploy at scale.

Our primary benchmark is Pavegen, the most established name in kinetic floor tiles; during our pilot we intend to run a direct side-by-side comparison showing that KineticStep's mechanical drivetrain is significantly cheaper to manufacture, install, and maintain than Pavegen's piezoelectric-based system, without compromising energy output.

The Second Revenue Stream: Data

Each tile can also be integrated with IoT sensors to collect anonymized data about pedestrian movement and footfall.

Shopping Malls and Commercial Spaces

Mall operators and individual stores can use footfall information to understand:

- Which areas receive the most traffic
- How people move through the building
- Which locations may be better for stores or advertising
- How customer movement changes over time

Metro Stations and Airports

Transport operators can use movement data to better understand passenger flow and support:

- Crowd management
- Queue planning
- Congestion analysis
- Infrastructure planning
- Operational decision-making

Importantly, the system is designed to provide these insights using anonymized movement data, without collecting personally identifiable information.

Where Can KineticStep Be Used?

Metro and Railway Stations

Tiles could be placed near ticket gates, entrances, platforms, and stairways to generate energy for lighting, displays, and sensors while providing insights into passenger movement.

Airports

High-traffic areas such as security queues, boarding gates, and arrival terminals provide opportunities for both energy generation and crowd-flow analytics.

Shopping Malls

KineticStep can act as both a sustainability feature and a source of valuable footfall insights for mall operators and individual retailers.

Universities and Smart Campuses

Campuses provide a strong environment for pilot installations, with high pedestrian movement and a growing focus on sustainability and smart infrastructure.

Bus Terminals and Other Public Spaces

The system can be used to support localized lighting, displays, sensors, and other low-power infrastructure in busy public areas.

Expected Energy Generation

The amount of energy generated depends on pedestrian traffic, tile efficiency, and mechanical performance. Current estimates suggest that one optimized tile in a high-footfall location could generate approximately 3–5 watts during peak traffic, with an estimated daily generation of around 0.1–0.3 kWh.

While the output from an individual tile is relatively small, the opportunity lies in large-scale deployment. A metro station, airport, or mall may have hundreds or thousands of tiles operating together. At this scale, the system can contribute meaningful localized energy while simultaneously generating valuable data about how the space is being used.

Our Progress

KineticStep has already taken several steps toward commercialization:

- ₹1 lakh grant received from NSUT for initial prototype development
- Incubation support from IGDTUW
- Provisional patent successfully filed for the technology
- Under evaluation for a potential ₹10 lakh grant through IIT Roorkee's TIDES NIDHI PRAYAS scheme

These milestones provide a foundation for moving from early-stage prototyping toward real-world validation.

Our Funding Ask

We are looking for funding to take KineticStep from a working proof-of-concept to an installation-ready product.

The funding will help us:

1. Refine and improve the existing prototype
2. Conduct testing for durability and long-term performance
3. Deploy pilot installations in real high-footfall locations
4. Benchmark KineticStep directly against existing solutions
5. Validate our cost and performance advantages
6. Develop the crowd analytics platform
7. Prepare the system for larger-scale manufacturing and deployment

Our long-term vision is to build a cost-effective and scalable smart infrastructure platform that combines two valuable outputs from everyday human movement: clean, localized energy and actionable pedestrian data.

KineticStep starts with something as simple as a person taking a step. Our goal is to turn those millions of everyday steps into a practical part of the sustainable, connected cities of the future.